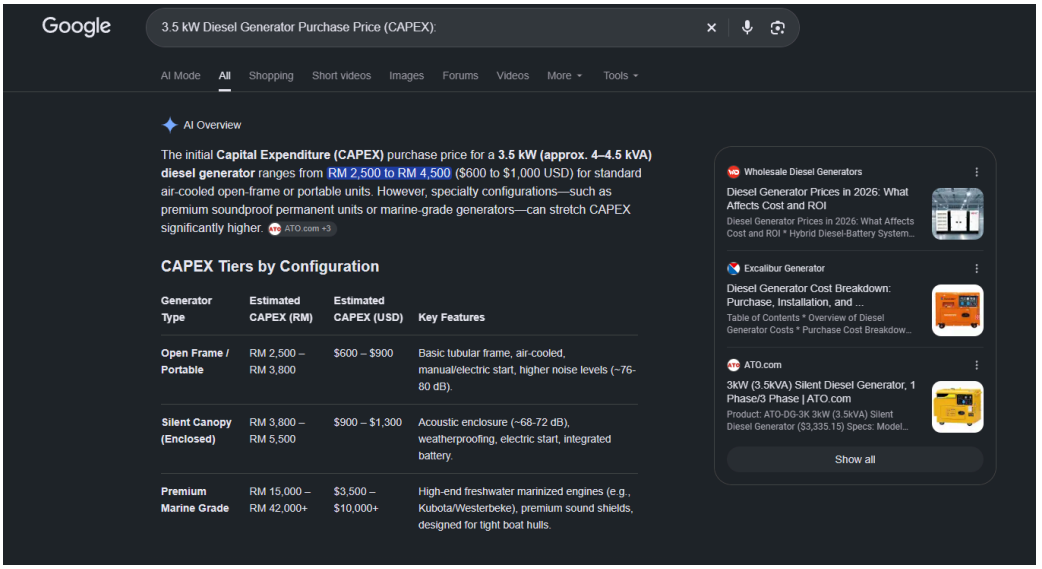
Reference 08: Diesel Generator Emissions & CAPEX

1. What is this? (Metrics & Checked Values)

- **Specific Fuel Consumption (SFC):** $0.30–0.33\ \mathrm{L/kWh}$ (Depco operational range: $0.20–0.40\ \mathrm{L/kWh}$ depending on load factor)
- **Diesel Emission Factor:** $2.687\ \mathrm{kg\ CO_2/L}$ (National Environment Agency Singapore / IPCC Standard)
- **Grid Carbon Intensity:** $0.80–0.90\ \mathrm{kg\ CO_2/kWh}$
- $3.5\ \mathrm{kW}$ **Standalone Diesel Generator CAPEX:** $\$800–\$1,200\ \mathrm{USD}$ (Commercial benchmark: ATO $3.5\ \mathrm{kW}$ silent generator $\approx \$1,050$)

2. The Proof (Screenshots & Technical Excerpts)

Domain	Screenshot Proof
Fuel Consumption & Efficiency	**How efficient are diesel generators and what influences fuel consumption?** Fuel efficiency is usually expressed as liters per kilowatthour (L/kWh) or liters per hour (L/h). Most diesel generators consume 0.2-0.4 L/kWh; for example, producing 50-kW while burning 10 liters per hour equates to 0.2 L/kWh. Larger machines use more fuel per hour but often deliver more kW per liter. At 75% load, typical consumption ranges from 4.5-5.5 L/h for a 20 kVA set to 150-165 L/h for a 1000 kVA unit. For context, these consumption rates align with figures from Cummins and Caterpillar for generators at optimal load. The fuel efficiency of a diesel generator improves when the generator operates at 70-80% of its rating. Oversizing causes wet stacking and wastes fuel; under sizing forces the engine to work harder and reduce efficiency. Implement proper load management (for example, shed nonessential loads or use automatic load balancing) to avoid underloading. Combined with right sizing and regular upkeep, these practices minimize fuel consumption.
Carbon Intensity Factor	Home / ESG Kopi Talk / CO2 per litre of diesel and petrol for company vehicles in Singapore (2.687 and 2.331 kg CO2e) **CO2 per litre of diesel and petrol for company vehicles in Singapore (2.687 and 2.331 kg CO2e)** ARTICLE · BY KAEL ONG · 8 OCT 2026 **Short answer:** burning one litre of **diesel** releases about **2.687 kg CO2e**, and one litre of **petrol** about **2.331 kg CO2e**. These are the factors we use for company vehicles in Singapore, built from the National Environment Agency's GHG Measurement & Reporting Guidelines (Appendix to Part II, Version 3). Multiply the litres in your fuel records by these numbers to get your Scope 1 emissions in kg CO2e.  Photo: Erik Mclean / Unsplash **Why "per litre", and where the number comes from** Your fuel-card statements and receipts show litres, so a per-litre factor lets you go straight from your records to emissions. NEA publishes its fuel factors per tonne of fuel and per unit of energy, not per litre, so we convert them in three steps:

Domain	Screenshot Proof
3.5 kW Generator CAPEX	

> **Depco Power Systems Fuel Consumption Guide:** Typical small diesel generators running at moderate loading (50%–75% load) consume 0.25–0.35 litres of diesel per kWh. At light loads ($< 25\%$, typical during night clinic hours), efficiency drops further towards 0.40 L/kWh.

> **Singapore NEA / UncleESG Carbon Reporting Standard:** Direct greenhouse gas combustion emission factor for stationary and automotive diesel fuel:
>
> $$\mathbf{2.687}\ \text{kg}\ \text{CO}_2\text{e per litre of diesel}$$

> **Commercial Equipment Price (ATO / Generic 3.5 kW Genset):** Market price for a standalone single-phase 3.5 kW (4.4 kVA) soundproof diesel genset ranges between $800 **and** $1,200 USD.

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3. How was it calculated?

- **Annual Fuel Consumption for $5,840$ kWh/year:**

$$\text{Fuel} = 5,840\ \text{kWh} \times 0.32\ \text{L/kWh} \approx \mathbf{1,868.8}\ \text{Litres/year}$$

- **Gross Diesel Emissions:**

$$\text{CO}_2 = 1,868.8\ \text{L} \times 2.687\ \text{kg CO}_2\text{/L} \approx \mathbf{5,021}\ \text{kg CO}_2\text{/yr} = \mathbf{5.02}\ \text{tonnes CO}_2\text{/year}$$

- **Net Avoided Carbon Emissions (Slide 9):**

$$\text{Net Avoided CO}_2 = 5.02\ \text{t (diesel)} - 0.27\ \text{t (solar)} = \mathbf{4.75}\ \text{tonnes CO}_2\text{/year}$$

$$\text{20-Year Lifetime Avoided} = 20 \times 4.75\ \text{t} = \mathbf{95.0}\ \text{tonnes CO}_2$$

- **Comparison to Generative AI Guess:**
  - AI guessed ~ 0.85 kg CO$_2$/kWh and ~ 4.96 tonnes/yr (99.2 tonnes over 20 years).
  - Grounded calculation (4.75 t/yr) proves the AI's emission reduction estimate was accurate within 4.2%.

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4. Source Link

- **Depco Fuel Consumption Guide:** Depco Diesel Generator Fuel Consumption Chart
- **NEA / Singapore ESG Factors:** UncleESG - CO2 Emissions per Litre of Diesel
- **ATO Generator Catalog:** ATO 3.5 kW Diesel Generator Catalog